

Samwel Waweru

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PROFESSIONAL SUMMARY

Senior AI Engineer with over four years of experience leading AI incubation and enterprise deployments across cloud, edge, and embedded environments. I take full technical ownership of cross functional programs, guiding work from requirements through proof of concept to production and adoption. I specialize in Generative AI, agent orchestration, and computer vision. I design scalable data pipelines, robust MLOps workflows, and reliable production inference services that deliver measurable productivity and business impact. I build and run experimentation frameworks, define KPIs for adoption and value, and translate technical outcomes into practical playbooks for non technical teams. I partner with senior stakeholders to align AI initiatives with strategy, coordinate cross functional advisors, and coach teams to adopt strong engineering and governance practices.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer (Kaggle BIPOC), Kaggle
Dec 2022 – Present

California, USA

- Lead the design, incubation, and delivery of production Generative AI services and agent orchestration platforms for internal and external stakeholders.
- Designed and evaluated Retrieval Augmented Generation solutions using FAISS and Pinecone, balancing accuracy, latency, and cost across experimentation and production phases.
- Built agent orchestration patterns that operationalize LLM driven workflows across cloud, edge, and IoT environments, enabling rapid prototyping and scalable deployment.
- Implemented containerized CI/CD pipelines, automated testing, and release workflows to ensure reproducible builds, safe experimentation, and reliable production rollouts.

AI Solutions Expert, EdTech Hub
Oct 2025 – Present

United Kingdom; Kenya (Hybrid)

- Lead cross functional AI initiatives from ideation through experimentation, technical validation, and production readiness for NLP and computer vision systems deployed at national scale.
- Established structured experimentation and pilot frameworks with clear success metrics, enabling evidence based decisions on AI adoption and scale.
- Delivered proof of concept and pilot systems with operational runbooks, monitoring dashboards, and formal handover processes to engineering, operations, and policy teams.
- Acted as a technical advisor to senior stakeholders, translating AI capabilities into practical implementation plans aligned with institutional priorities and productivity outcomes.

County Coordinator, Raspberry Pi Foundation
Apr 2024 – Oct 2025

United Kingdom; Kenya (Hybrid)

- Established ten Code Clubs and trained over 250 learners in Python, embedded systems, and AI fundamentals, driving grassroots adoption of applied computing and AI skills.
- Designed and delivered structured training programs, workshops, and hands on sessions to support experimentation, learning, and measurable skill progression.
- Coordinated educators, nonprofits, and local government stakeholders to align digital learning and AI enablement initiatives with regional development and workforce priorities.
- Built and sustained a cross functional community of practice, translating technical concepts into accessible playbooks and learning pathways for non specialists.

EDUCATION

Jomo Kenyatta University of Agriculture and Technology
B.Sc. Mechatronic Engineering, Robotics - Honors (2:1)

Nairobi, Kenya

- **Capstone Project (Rank 1/18):** Designed a Delta Laser-Weeding Robot with computer vision, real-time inference (Raspberry Pi + YOLOv11), and distributed microcontroller orchestration.
- Took courses on Calculus 1- 4, Advanced Robotics, Differential Equations, and Computer Science.

The University of Manchester
Summer School - Machine Learning, Big Data

United Kingdom, Manchester
Aug 2021

- Built and deployed NLP models using Hugging Face Transformers (BERT) for sentiment analysis of Twitter data via API integration (Python).
- **Hackathon Winner:** Authored award-winning report on “Effects of COVID-19 vis-à-vis Mental Health”, applying data-driven modeling and analytics to a real-world challenge.

PUBLISHED PAPERS

- **Samwel Waweru** (Lead Author). *ML-Enhanced Fellegi-Sunter for Data Validation and Preparation*. Manuscript in preparation. Planned submission to NeurIPS 2027. The paper integrates machine learning based feature engineering and calibration into the Fellegi-Sunter framework to improve automated data validation and record linkage workflows.
- **Samwel Waweru** (Lead Author), Ray Njoroge. *Edge Optimized YOLOv11 Based Real Time Weed Detection for Autonomous Precision Agriculture*. Accepted at IEEE RSJ IROS 2025. The work presents a production ready edge perception pipeline with optimized inference, data driven evaluation and deterministic integration with autonomous actuation systems under real world latency and power constraints.
- Jackson Mutembei, **Samwel Waweru** (Contributor). *A Biodiversity Conservation Protocol for Fire Prevention using Drone Technology*. University of Birmingham, 2022. I contributed to the design of an end to end drone based monitoring pipeline combining computer vision, sensor fusion and data validation workflows to support early warning and decision support systems.

ENGINEERING PROJECTS

- **Delta Laser Weeding Robot** (Computer Vision + Robotics Backend) Built a computer vision inference service around **YOLOv11** running on Raspberry Pi 4, optimizing it using INT8 quantization and I/O pipeline tuning to reduce processing latency. Designed Python modules that expose detection results through structured interfaces consumed by the robot’s motion controllers. Debugged and maintained data flow between the CV service, G-code generator, and microcontroller firmware to ensure stable field operation. (*Key Tech: Python, YOLOv11, TensorFlow, OpenCV, REST-like Module Interfaces, Raspberry Pi, Microcontroller Integration.*) [Link](#)
- **EmbedMind Studio** I developed a lightweight platform for assembling and deploying LLM powered agents on embedded devices. I implemented a Python backend for agent orchestration, I integrated with LLM APIs and I managed device communication over serial and Wi Fi. I used Docker to create reproducible development and simulation environments and I implemented CI CD workflows to ensure consistent builds across devices. (*Key Tech: Python, C++, Docker, CI CD Workflows, LLM APIs, Embedded Networking.*) [Link](#)
- **Golden Record for KEMIS** I designed an AI enabled record linkage system to resolve duplicate and inconsistent learner and school records. I built scalable ingestion and normalization pipelines and I combined rule based entity resolution with similarity scoring and human in the loop validation to provide auditable trails for governance. (*Key Tech: Python, Entity Resolution, Fuzzy Matching, Data Pipelines, Human in the Loop Systems.*)
- **MentalMuse** Created a secure NLP analysis pipeline for mental-health text. Implemented scalable data-processing services using Python and orchestrated batch/stream workloads using **Snowpark**. Built API integrations to route users to mental-health support services and added lightweight explainability features around NLP outputs. Focused on stable data handling, schema consistency, and secure text processing. (*Key Tech: Python, NLP Pipelines, Snowpark (Snowflake), Secure API Integration, Data Validation.*) [Link](#)

- **Kilimo-AI** Built a simple LLM-assisted irrigation control system for IoT hardware. Created Python-C++ modules to process user natural-language input, apply prompt-engineering logic, and translate decisions into device-level irrigation commands. Integrated cloud REST APIs for device state updates and alerts. Designed the system with low-power hardware constraints in mind to ensure stable operation in rural settings. (*Key Tech: Python, C++, LLM Prompt Engineering, IoT Controllers, REST APIs, Cloud Integration.*) [Link](#)

PROFESSIONAL CERTIFICATIONS

- Oracle Cloud Infrastructure Certified Data Science Professional. Credential ID: OC4504520.
- AWS Certified Generative AI Professional, with focused study on production Generative AI architectures, foundation model integration, security, and cost optimized deployment patterns.

SKILLS AND INTERESTS

- **Generative AI and Agent Orchestration:** Design, incubation, and production deployment of Generative AI systems and agent based workflows using PyTorch, LangChain, and major LLM APIs, with a focus on reliability, governance, and measurable productivity gains.
- **MLOps and AI Platform Development:** End to end MLOps for experimentation and production, including CI CD for models, reproducible container builds, model registry patterns, automated validation, monitoring, and lifecycle management.
- **Data Engineering and Governance:** Design of scalable ingestion and transformation pipelines with strong auditability, data quality controls, and compliance for enterprise and public sector systems.
- **Computer Vision and Applied Perception:** Development and deployment of vision systems for edge environments, including dataset benchmarking, model optimization, and quantization to meet latency, reliability, and operational constraints.
- **Cloud and Infrastructure Architecture:** Cloud native architecture patterns on AWS and Azure, with Git based deployment workflows, cost awareness, and secure production operations.
- **Embedded Systems and Edge AI:** C and C++ development for embedded and IoT systems, including CAN, Bluetooth, GPS, and networked devices, enabling AI deployment beyond the cloud.
- **Software Engineering and Delivery Practices:** Clean code, modular design, automated testing, DevOps practices, and cross functional collaboration to support scalable and maintainable systems.
- **Languages:** English fluent; Kiswahili native; Japanese basic.